14 White shapes and black shapes are used in a game.
Some of the shapes are circles.
All the other shapes are squares.

The ratio of the number of white shapes to the number of black shapes is 3 : 7

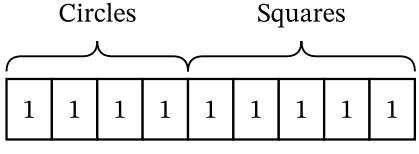
The ratio of the number of white circles to the number of white squares is 4 : 5

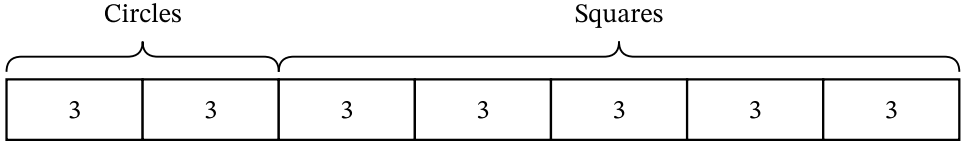
The ratio of the number of black circles to the number of black squares is 2 : 5

Work out what fraction of all the shapes are circles.

Choose 30 shapes: white : black = 9 : 21.

White has 9 parts; black has 7 parts of 3.





White circles = 4 × 1 = 4

1 mark
Find white circles

Black circles = 2 × 3 = 6

1 mark
Find black circles

Total circles = 4 + 6 = 10 out of 30.

1 mark
Combine both circle counts

Fraction = $\frac{10}{30} = \frac{1}{3}$

1 mark
Correct fraction

$\frac{1}{3}$

(Total for Question 14 is 4 marks)